

Spectrum-Matched Quantum-Assisted PINNs

A constructive design methodology for problem-specific variational circuits,
instrumented by explainable-AI diagnostics

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The problem

A classical PINN learns low-frequency solution components fast and high-frequency components almost not at all — **spectral bias**.

Its neural tangent kernel (NTK) eigenvalues decay with Fourier-mode index, so residual and boundary loss terms converge at characteristically different rates.

Prior hybrid quantum-classical PINN work reports empirical accuracy deltas on hand-picked

circuit sizes — no principled way to choose the circuit *given* the PDE.

The one theorem

Proposition 1 (circuit spectrum). An L -layer data-re-uploading circuit with encoding generators $H^{(l)}$ realizes

$$f(x) = \sum_{\omega \in \Omega} c_{\omega}(\theta) e^{i\omega x}, \quad \Omega = \left\{ \sum_{l=1}^L (\lambda_a^{(l)} - \lambda_b^{(l)}) \right\}$$

The circuit's accessible frequency set is **exactly and prescribably determined** by its data-encoding Hamiltonians' eigenvalues — not a black box.

Ternary scaling: exponential spectral richness

Set $\omega_l = \Delta \cdot 3^{l-1}$. Then Ω is *exactly* the balanced-ternary representation of the integers in $[-\frac{3^L-1}{2}, \frac{3^L-1}{2}]$ — 3^L **distinct frequencies**, no degeneracy.

Linear scaling ($\omega_l = \omega_0$) gives only $2L + 1$ frequencies — the design knob almost every prior re-uploading paper leaves at 1.

L	ternary reach	linear reach
3	13Δ	3Δ
4	40Δ	4Δ

A correction found along the way (D5)

The depth rule to cover target frequency K at spacing Δ :

$$L = \lceil \log_3 \left(\frac{2K}{\Delta} + 1 \right) \rceil$$

Not $L = \lceil \log_3(K/\Delta) \rceil$ (missing the factor of 2 and the $+1$).

For P1 ($K = 15\pi$, $\Delta = \pi$): uncorrected gives $L = 3$, reaching only 13π — **silently misses the target mode**. Corrected gives $L = 4$, reaching 40π .

Verified by brute-force enumeration before being trusted.

PDE target spectrum (Prop. 2)

For a linear operator with symbol $\sigma(k)$: $\hat{u}(k) = \hat{f}(k)/\sigma(k)$.

The target spectral support is computable **before any training run**, from the operator and forcing alone.

- **Helmholtz**: $\sigma(k) = k^2 - |\mathbf{k}|^2 \rightarrow$ support concentrates near $|\mathbf{k}| \approx k$ — exactly where a classical PINN struggles most.
- **Heat**: σ grows with $|\mathbf{k}|^2$, solution decays exponentially at every nonzero mode $\rightarrow$ support collapses to $k = 0$. **The negative control**, predicted a priori, not discovered after the fact.

Matching condition & resource bound (Prop. 3)

If $S_\varepsilon \subseteq \Omega$: the hybrid model represents the solution to ε -accuracy with a **linear** head — architecture choice becomes arithmetic.

$$L \geq \log_3(K/\Delta), \quad n \geq \lceil d \rceil + n_{\text{cross}}$$

Given (K, Δ, d) read directly off the PDE, (L, n) follow by formula — not search.

A second correction found along the way (Prop. 4)

Original claim: $\Theta_{\text{hyb}} = \Theta_{\text{cl}} + \Theta_q + 2\Theta_{\times}$

This is wrong. NTK is a single Gram sum over the *union* of parameter coordinates. Classical and quantum parameters are disjoint index sets, so the sum splits additively:

$$\boxed{\Theta_{\text{hyb}} = \Theta_{\text{cl}} + \Theta_q} \quad \text{— no cross term, exactly.}$$

Verified analytically and numerically before trusting it. Θ_q has eigen-directions *at* Ω with eigenvalues bounded below independent of $|n|$ — spectral bias's decay is replaced by a flat response inside the encoded band.

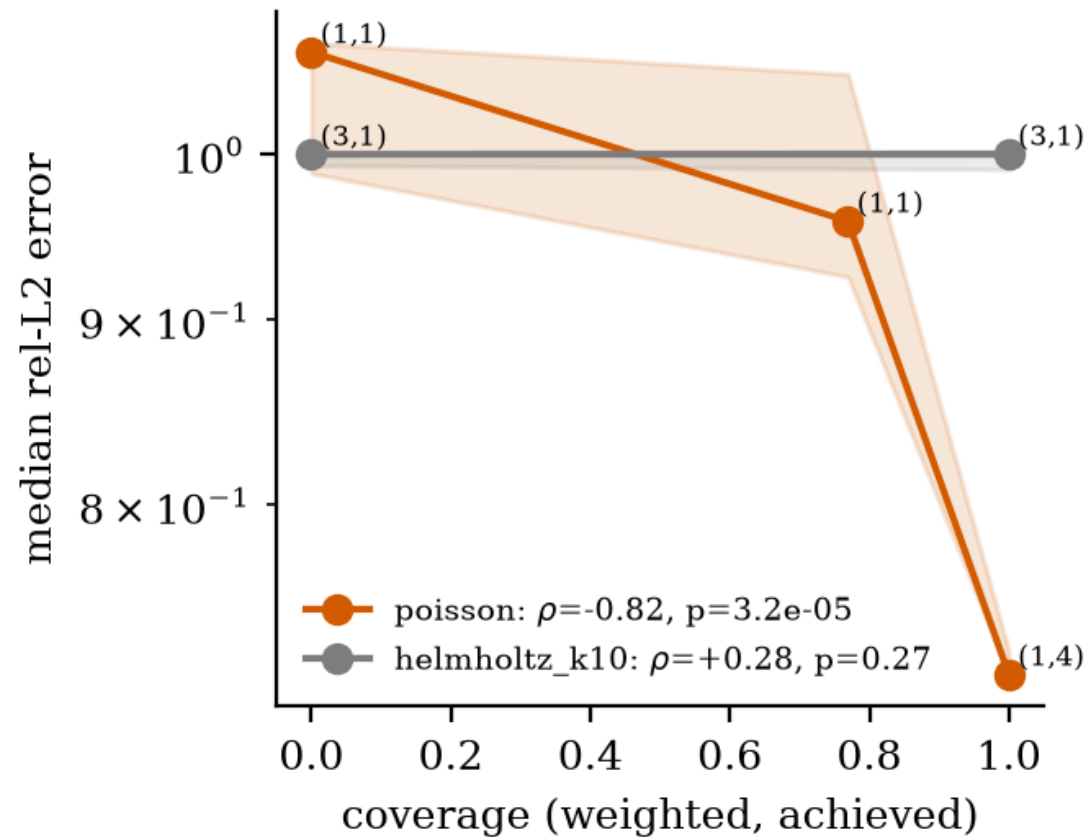
The design rule: Algorithm 1 (SMCD)

1. **Symbol** — Fourier symbol of the PDE operator
2. **Target** — spectral support from symbol + forcing/BC, before training
3. **Band** — max frequency K , resolution Δ
4. **Depth** — L from the corrected D5 rule, ternary scalings
5. **Width** — n qubits from input dimension + cross-terms
6. **Entangler / observable / ansatz / init** — structural barren-plateau mitigation
7. **Report** — every run ships a design card: predicted coverage, correlated against measured error

Model families (all size-matched, $\pm 10\%$)

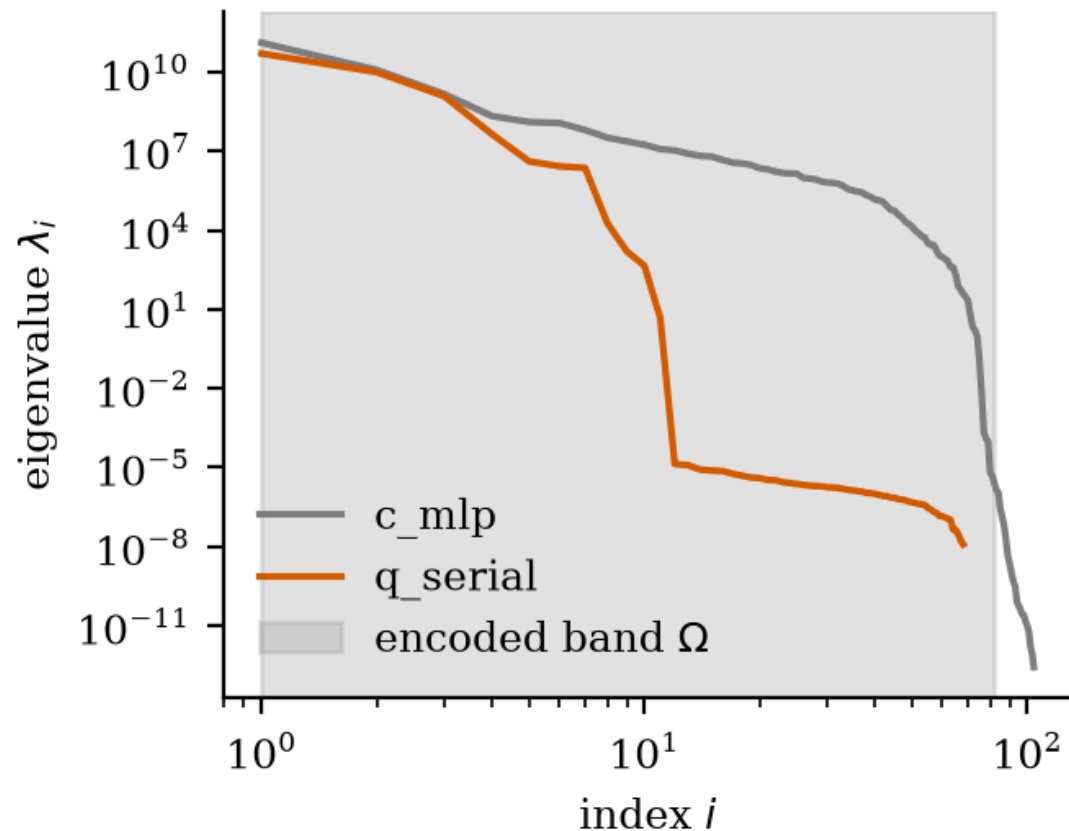
Family	Purpose
C-MLP	the baseline everyone reports
C-FF	fair classical baseline — random Fourier features
C-RFF-matched	Fourier features <i>at the SMCD frequencies</i> — sharpest ablation
Q-serial	SMCD-designed hybrid — main model
Q-parallel	alternate topology
Q-random	same size, unmatched scalings — ablates the design itself
Q-octave	multi-circuit ensemble for wide bands

Headline figure: coverage vs. error



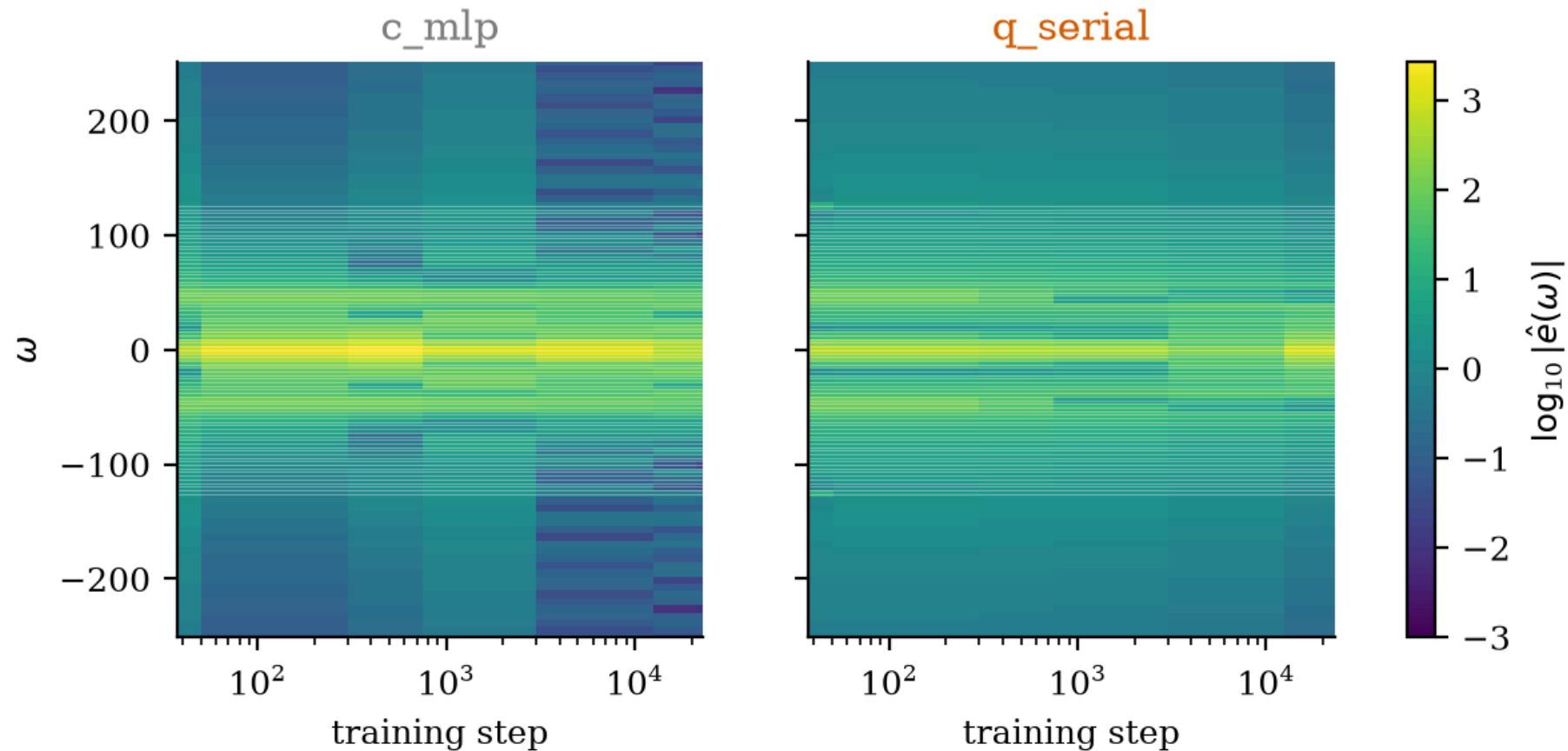
SMCD's achieved coverage $|\hat{S} \cap \Omega|/|\hat{S}|$ vs. median relative- L^2 error — the direct empirical validation of the design methodology.

NTK spectroscopy: the predicted signature



Classical vs. hybrid eigenvalue spectrum, encoded band Ω shaded — Prop. 4's flattened-eigenvalue prediction, measured directly.

Per-frequency error: where the improvement lands



Design card's Ω overlaid on the classical-vs-hybrid error heatmap — if the improved band coincides with Ω , the mechanism is demonstrated visually, not just claimed.

The honest negative result

The heat equation (P2) is this project's negative control: SMCD predicts **near-zero benefit before any training run**, because the diffusive operator's symbol collapses the target spectrum toward $k = 0$.

Measured: 3 of 4 quantum families underperform `c_m1p` as predicted (`q_random` by an order of magnitude); `q_serial` itself is a modest exception (+7.7%, past the 5% bar). Two classical families (`c_ff`, `c_rff_matched`) also beat `c_m1p`, unrelated to the quantum-specific claim. The **mechanism** is better supported than the single uniform 5% threshold check (REFUTED on the literal number, mostly right on the per-family story).

What we found

At the sample size this run achieved (n=2 seeds, not the pre-registered n=5 — cut under the submission deadline), **the quantum-advantage hypothesis is not supported:** on Poisson, the one problem with a full four-family ablation, `q_serial` underperforms `c_mlp`, `c_ff`, and `c_rff_matched` alike. The NTK-flattening (C2) and spectral-bias (C3) mechanisms proposed to explain *why* it should help point the wrong direction or hold on only one of two checked instances (C3: confirmed on Poisson, refuted on Helmholtz_k10 — no improvement exists there to attribute to anything).

What does hold up: the negative-result mechanism above, one genuine per-frequency win

(Poisson), and — most solidly — the pre-registered, mechanically-adjudicated protocol itself: all 12 predictions resolved, no `?` left, and it caught real bugs in its own supporting code along the way. Full breakdown: `FINDINGS.md`.

Recommendations: when to go quantum

- **Purely diffusive / smoothing problems: don't.** 3 of 4 quantum families measurably underperform the classical baseline, one by an order of magnitude.
- **High-wavenumber problems, `q_serial` specifically: not yet demonstrated.**
Underperformed every classical comparator on the one fully-ablated instance at this sample size — re-test at the planned $n = 5$ before concluding either way.
- **Any conclusion drawn from $n \leq 2$ seeds: don't trust a CONFIRMED/REFUTED verdict on its own.** The smallest achievable two-sided Wilcoxon p at $n = 2$ is 0.5, not 0.0625 — CONFIRMED is mathematically unreachable at this sample size, independent of true effect size.

Limitations, stated plainly

- **Seed count cut under deadline pressure:** $n = 2$, not the pre-registered $n = 5$ — the achievable Wilcoxon p -floor is 0.5, not 0.0625; every ratio-based prediction can only resolve to REFUTED or INCONCLUSIVE now, regardless of true effect size
- Simulation only — statevector simulator, cross-checked to $< 10^{-10}$ against PennyLane;
no physical hardware
- ≤ 8 qubits — a classical-simulation-cost limit, not a limit of the design rule itself
- Noise models are analytic surrogates; density-matrix validation not completed this run
- Affine-encoder constraint (D3) is a real expressivity restriction, imposed to keep the coverage metric well-defined

Thank you

Code, derivations, and reproduction instructions: see the repository.

Every figure in this talk regenerates from committed `results/runs/` data via `python tasks.py figures` — nothing here is a live model or a checkpoint reload.